

Alexandria University
Faculty of Engineering
Mechanical Engineering Department

Signal Analysis

Ahmed Mohamed Ahmed Morsi Bsher

1. ***The resultant of two harmonics with different amplitudes and the same frequency.***

Discerption: The resultant is a single harmonic wave. Because the frequencies are identical, the two signals simply add together to form one pure sine wave with a combined amplitude. The frequency spectrum shows a single, isolated peak at that exact frequency.

Case 1 : MATLAB Code

```
% Case 1: Two harmonics, different amplitudes, same frequency
dt = 0.005;
t = 0:dt:20;

A1 = 3;
A2 = 1.5;
w1 = 20;
w2 = 20;

x1 = A1*sin(w1*t);
x2 = A2*sin(w2*t);
x = x1 + x2;

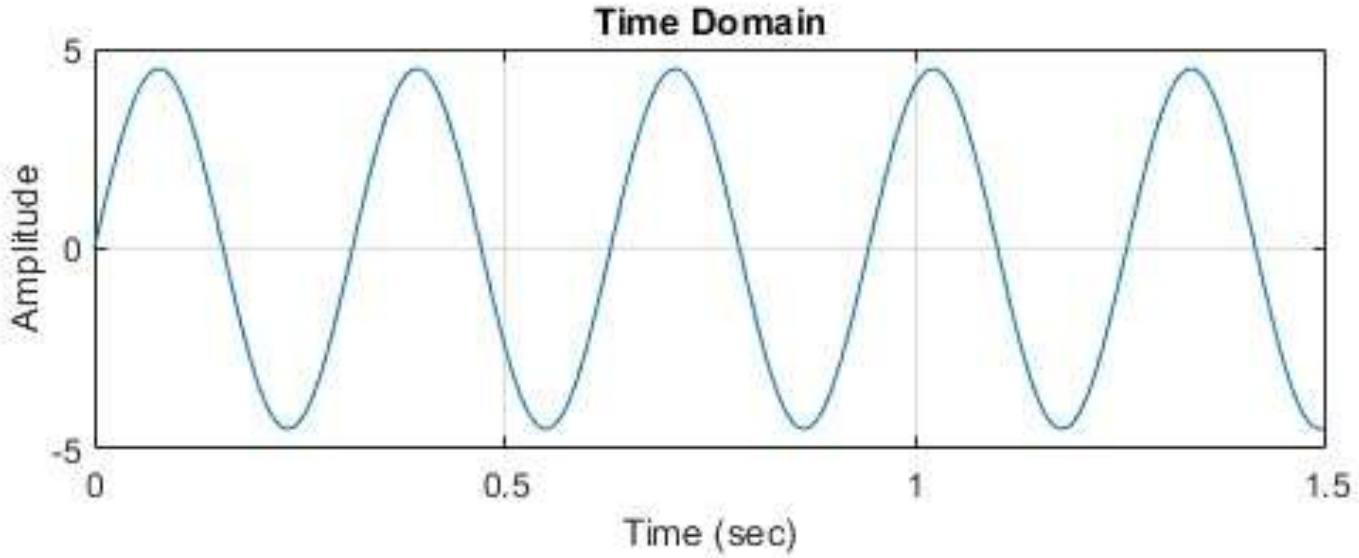
subplot(2,1,1)
plot(t, x)
grid on
xlabel('Time (sec)')
ylabel('Amplitude')
title('Time Domain')
axis([0 1.5 -5 5])

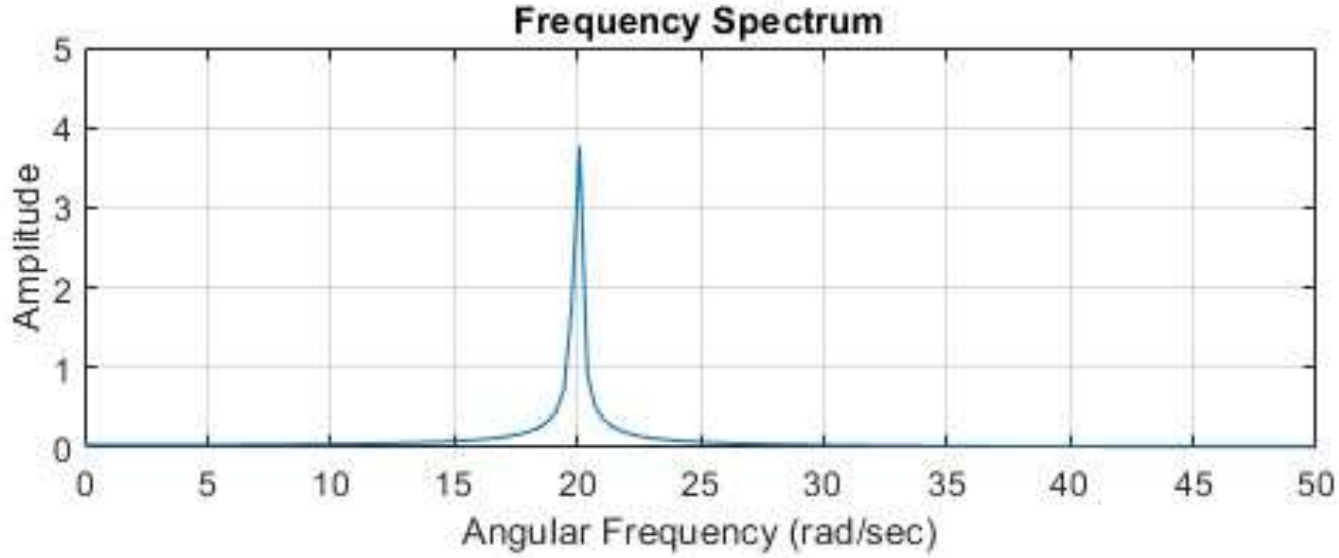
% FFT Calculation
fs = 1/dt;
N = length(t);
f = (0:N-1)*(fs/N);
w_axis = f * 2 * pi; % Convert Hz to rad/sec

xt = fft(x);
xf = (abs(xt)*2)/N;

subplot(2,1,2)
plot(w_axis, xf)
grid on
xlabel('Angular Frequency (rad/sec)')
ylabel('Amplitude')
title('Frequency Spectrum')
axis([0 50 0 5]) % Zoom in on the relevant frequency range
```

Case 1 : Results





2. *The resultant of two harmonics with equal amplitudes, where the first frequency is very close to the second one ($\omega_1 \approx \omega_2$).*

Discerption: This combination produces the "beat" phenomenon. The slight difference in frequencies causes the signals to continuously shift in and out of phase, alternating between constructive and destructive interference. This creates a fast carrier wave trapped inside a slowly pulsating amplitude envelope. The frequency spectrum shows two distinct, closely spaced peaks.

Case 2：MATLAB Code

```
% Case 2: Two harmonics, equal amplitudes, very close frequencies
dt = 0.005;
t = 0:dt:20;

A1 = 2;
A2 = 2;
w1 = 50;
w2 = 52;

x1 = A1*sin(w1*t);
x2 = A2*sin(w2*t);
x = x1 + x2;

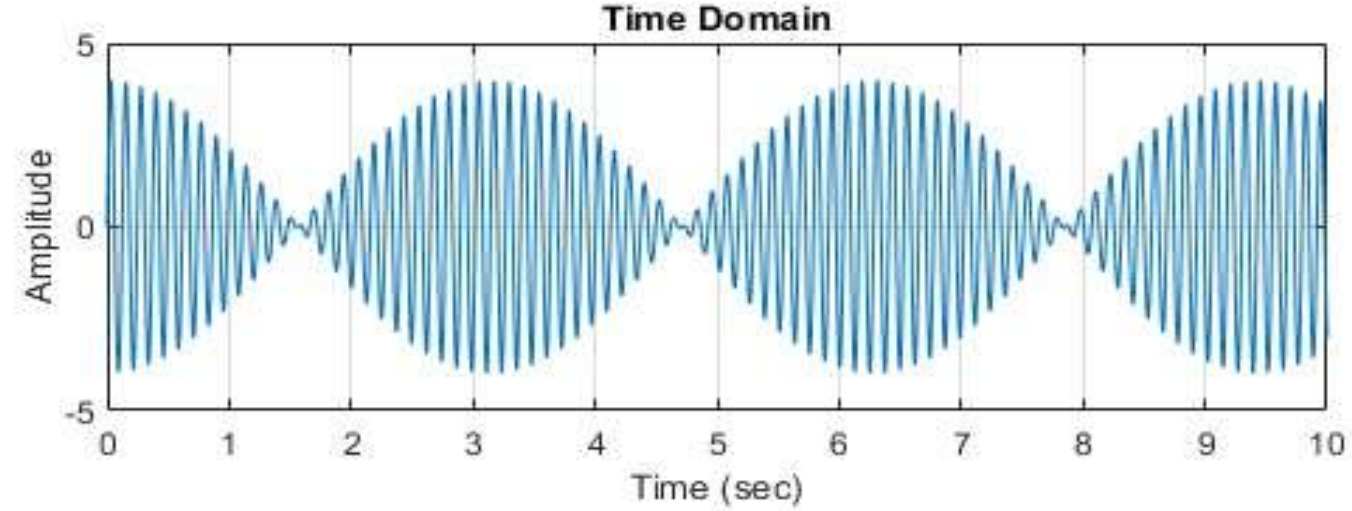
subplot(2,1,1)
plot(t, x)
grid on
xlabel('Time (sec)')
ylabel('Amplitude')
title('Time Domain')
axis([0 10 -5 5])

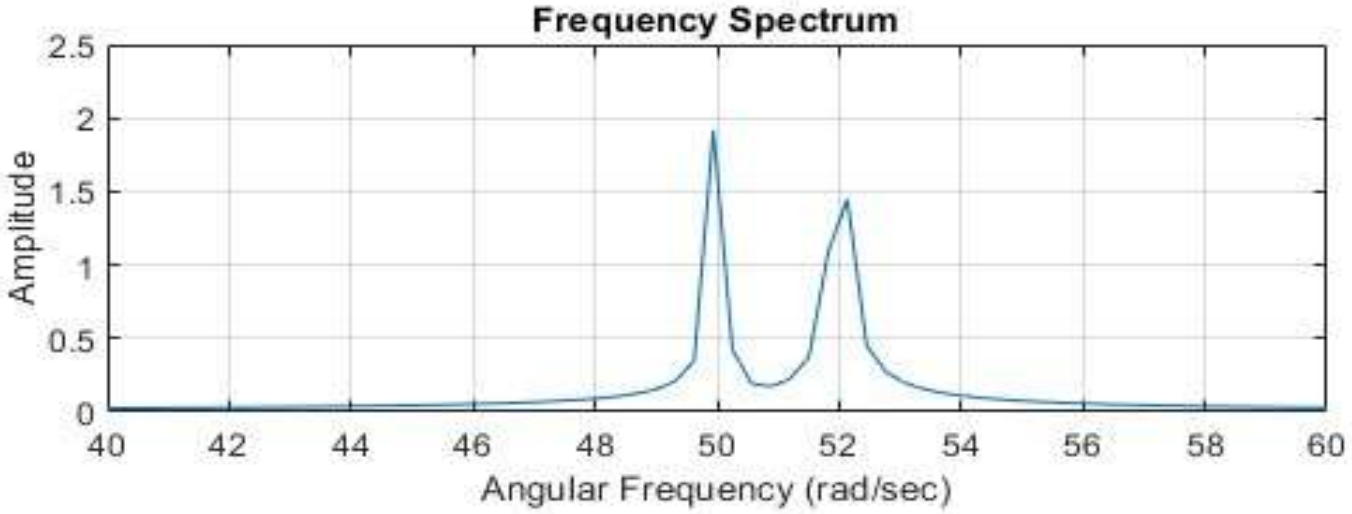
% FFT Calculation
fs = 1/dt;
N = length(t);
f = (0:N-1)*(fs/N);
w_axis = f * 2 * pi;

xt = fft(x);
xf = (abs(xt)*2)/N;

subplot(2,1,2)
plot(w_axis, xf)
grid on
xlabel('Angular Frequency (rad/sec)')
ylabel('Amplitude')
title('Frequency Spectrum')
axis([40 60 0 2.5])
```

Case 2 : Results





3. *The resultant of two harmonics with different amplitudes and widely separated frequencies ($\omega_1 \gg \omega_2$).*

Discerption: The resultant signal consists of a fast, small-amplitude ripple superimposed on top of a slow, dominant wave. Because the frequencies are vastly different, they do not blend into a beat pattern. The frequency spectrum displays two completely separate peaks located far apart on the axis.

Case 3 : MATLAB Code

```
% Case 3: Two harmonics, different amplitudes, widely separated frequencies
dt = 0.005;
t = 0:dt:20;

A1 = 5;
A2 = 1.5;
w1 = 5;
w2 = 100;

x1 = A1*sin(w1*t);
x2 = A2*sin(w2*t);
x = x1 + x2;

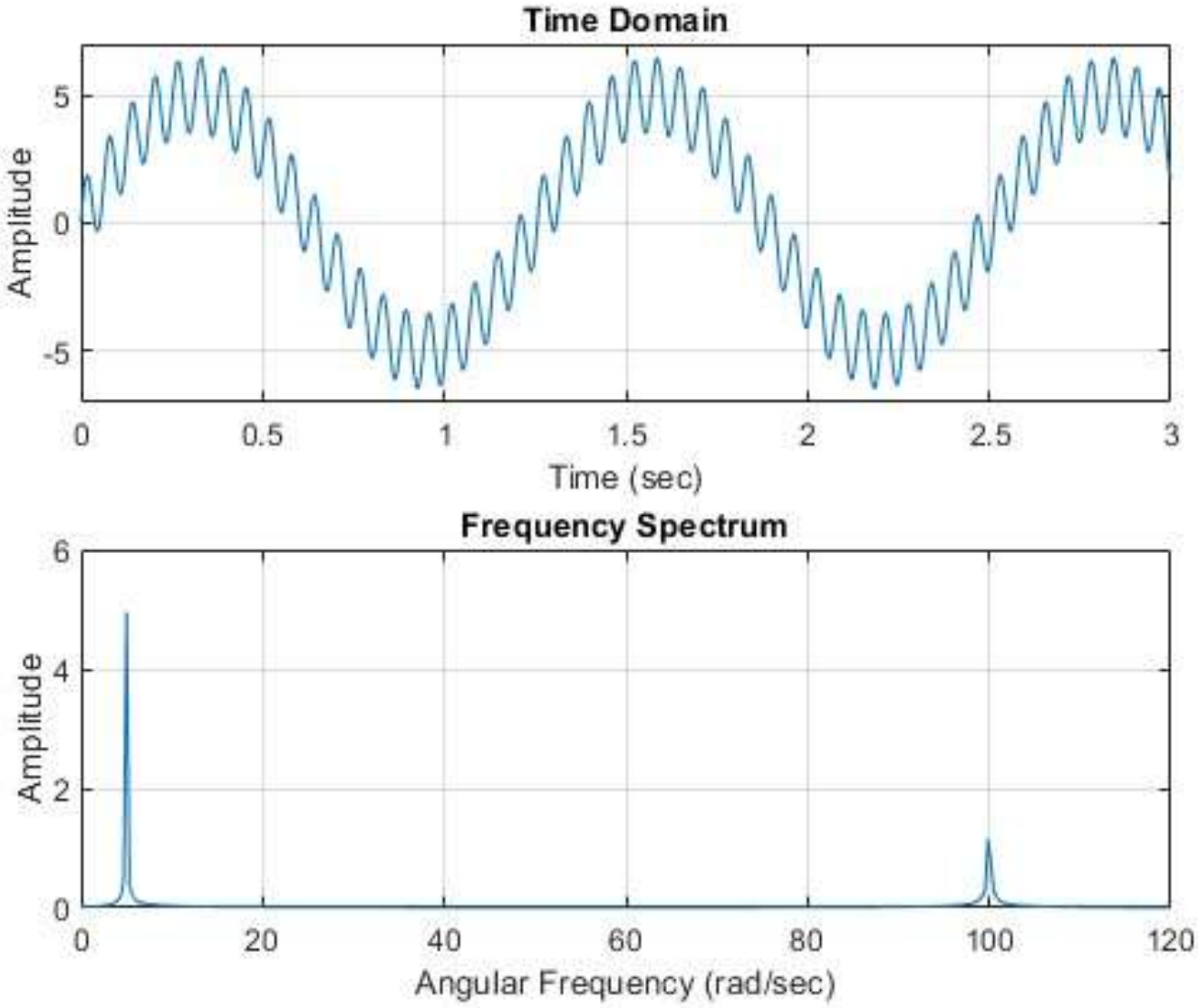
subplot(2,1,1)
plot(t, x)
grid on
xlabel('Time (sec)')
ylabel('Amplitude')
title('Time Domain')
axis([0 3 -7 7])

% FFT Calculation
fs = 1/dt;
N = length(t);
f = (0:N-1)*(fs/N);
w_axis = f * 2 * pi;

xt = fft(x);
xf = (abs(xt)*2)/N;

subplot(2,1,2)
plot(w_axis, xf)
grid on
xlabel('Angular Frequency (rad/sec)')
ylabel('Amplitude')
title('Frequency Spectrum')
axis([0 120 0 6])
```

Case 3 : Results



4. ***The resultant of 10 harmonics with equal amplitudes and different frequencies as follows: $\omega_1 = 10$, $\omega_2 = 20$, $\omega_3 = 30$, … , $\omega_{10} = 100$ (rad/sec).***

Discerption: The addition of these evenly spaced harmonics: -

$$(\omega = 10, 20, \dots, 100\ rad/s)$$

creates a complex but strictly periodic non-sinusoidal waveform. Because all amplitudes are equal, the high-frequency components contribute just as much energy as the low-frequency ones. The frequency spectrum displays ten uniform peaks spaced exactly 10 rad/s apart.

Case 4 : MATLAB Code

```
% Case 4: 10 harmonics, equal amplitudes, w = 10, 20... 100 rad/s
dt = 0.005;
t = 0:dt:20;
N_samples = length(t);

x = zeros(1, N_samples);
A = 2;

% Simple loop to add the 10 harmonics
for i = 1:10
    w = i * 10;
    x = x + A*sin(w*t);
end

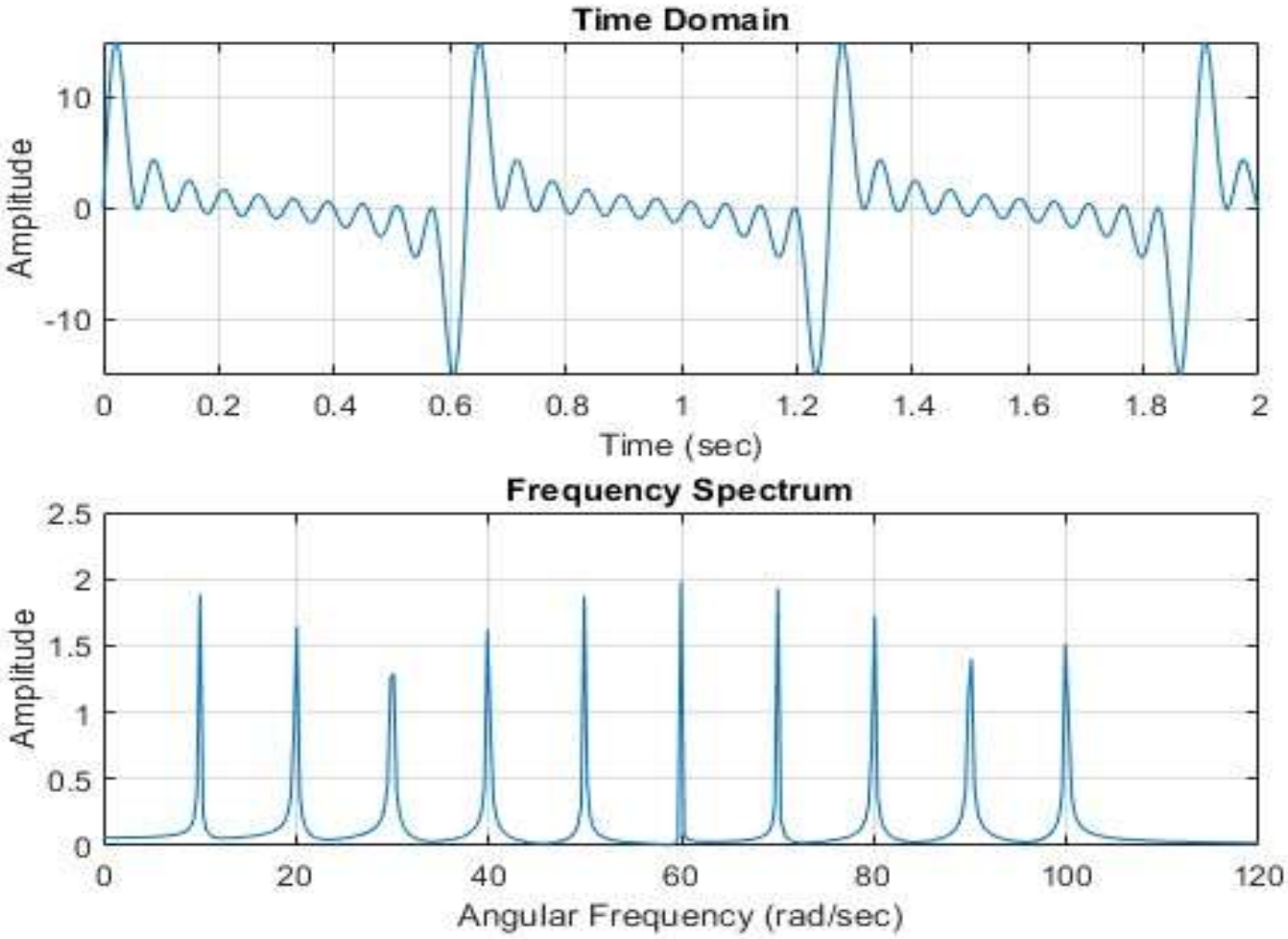
subplot(2,1,1)
plot(t, x)
grid on
xlabel('Time (sec)')
ylabel('Amplitude')
title('Time Domain')
axis([0 2 -15 15])

% FFT Calculation
fs = 1/dt;
f = (0:N_samples-1)*(fs/N_samples);
w_axis = f * 2 * pi;

xt = fft(x);
xf = (abs(xt)*2)/N_samples;

subplot(2,1,2)
plot(w_axis, xf)
grid on
xlabel('Angular Frequency (rad/sec)')
ylabel('Amplitude')
title('Frequency Spectrum')
axis([0 120 0 2.5])
```

Case 4 : Results



5. ***The resultant of 10 harmonics with different amplitudes and different frequencies as follows: $\omega_1 = 10$, $\omega_2 = 20$, $\omega_3 = 30$, … , $\omega_{10} = 100$ (rad/sec).***

Discerption: While the frequencies are identical to Case 4, the decreasing amplitudes mean the higher frequencies carry progressively less energy. This results in a visually smoother periodic waveform in the time domain. The frequency spectrum shows ten evenly spaced peaks forming a linear downward staircase pattern.

Case 5：MATLAB Code

```
% Case 5: 10 harmonics, different amplitudes, w = 10, 20... 100 rad/s
dt = 0.005;
t = 0:dt:20;
N_samples = length(t);

x = zeros(1, N_samples);

% Loop to add harmonics with decreasing amplitudes
for i = 1:10
    w = i * 10;
    A = 11 - i; % Amplitudes will be 10, 9, 8... 1
    x = x + A*sin(w*t);
end

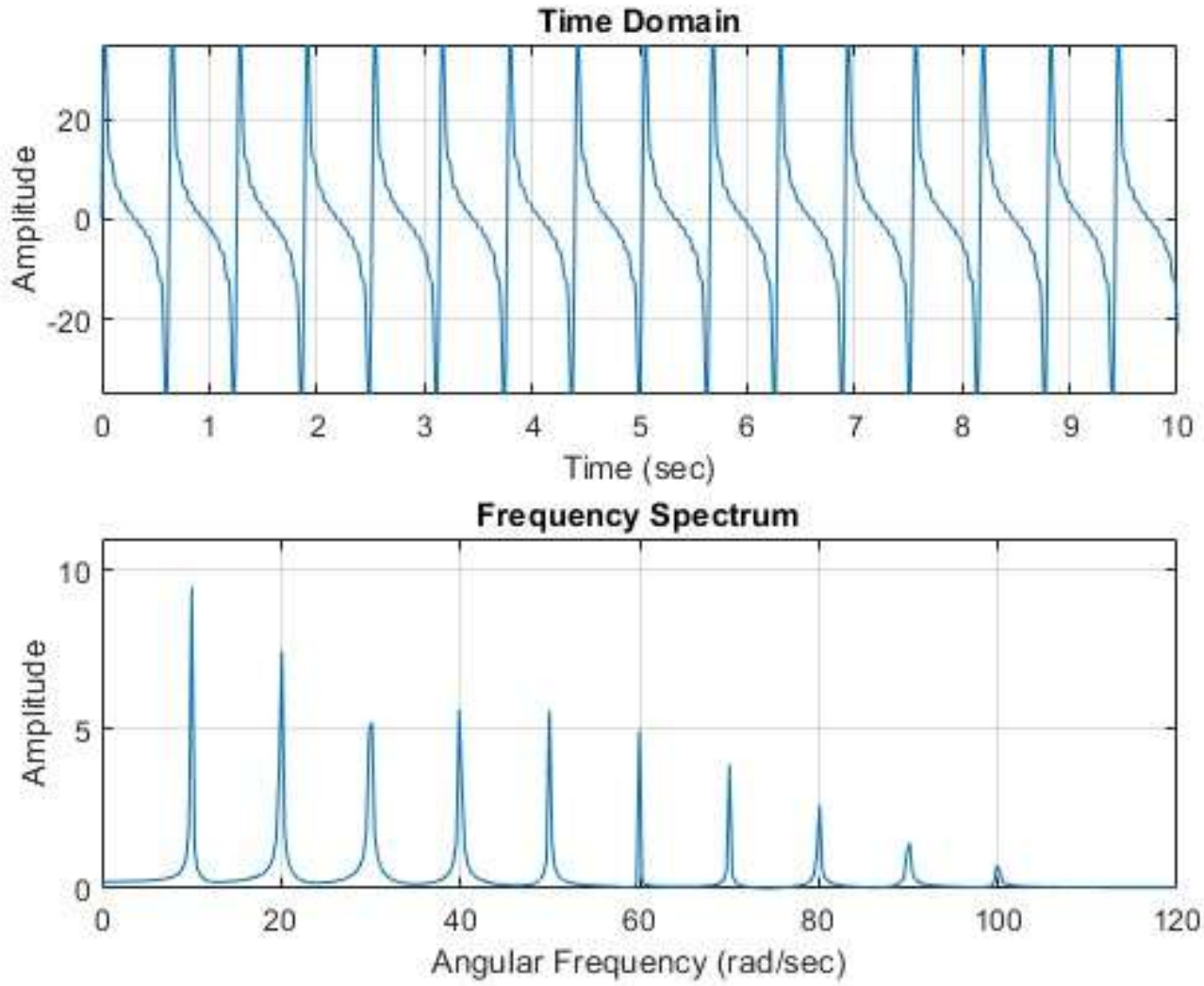
subplot(2,1,1)
plot(t, x)
grid on
xlabel('Time (sec)')
ylabel('Amplitude')
title('Time Domain')
axis([0 2 -35 35])

% FFT Calculation
fs = 1/dt;
f = (0:N_samples-1)*(fs/N_samples);
w_axis = f * 2 * pi;

xt = fft(x);
xf = (abs(xt)*2)/N_samples;

subplot(2,1,2)
plot(w_axis, xf)
grid on
xlabel('Angular Frequency (rad/sec)')
ylabel('Amplitude')
title('Frequency Spectrum')
axis([0 120 0 11])
```

Case 5 : Results



6. *The resultant of 10 harmonics with different amplitudes and randomly chosen frequencies.*

Discerption: The resultant is a chaotic, aperiodic signal. Because the frequencies are chosen randomly, they lack a common fundamental period and do not perfectly align to form a repeating pattern. The frequency spectrum reveals ten scattered peaks of varying heights placed irregularly along the frequency axis.

Case 6：MATLAB Code

```
% Case 6: 10 harmonics, random amplitudes and frequencies
dt = 0.005;
t = 0:dt:20;
N_samples = length(t);

x = zeros(1, N_samples);
rng(1); % Seed for reproducible random numbers

for i = 1:10
    % Random w between 10 and 200, Random A between 1 and 5
    w = 10 + rand() * 190;
    A = 1 + rand() * 4;
    x = x + A*sin(w*t);
end

subplot(2,1,1)
plot(t, x)
grid on
xlabel('Time (sec)')
ylabel('Amplitude')
title('Time Domain')
axis([0 3 -25 25])

% FFT Calculation
fs = 1/dt;
f = (0:N_samples-1)*(fs/N_samples);
w_axis = f * 2 * pi;

xt = fft(x);
xf = (abs(xt)*2)/N_samples;

subplot(2,1,2)
plot(w_axis, xf)
grid on
xlabel('Angular Frequency (rad/sec)')
ylabel('Amplitude')
title('Frequency Spectrum')
axis([0 220 0 6])
```

Case 6 : Results

